

Sequences

9 March 2024

Problem 1. Prove that if an arithmetic sequence of positive integers contains a square, then it contains infinitely many squares.

Solution. Assume with no loss of generality that the first term is a square, say a^2 , and let d be the common difference. Hence the sequence is $a^2, a^2 + d, a^2 + 2d, \dots$. Eventually we get to $a^2 + (2a + d)d = (a + d)^2$. By repeating the argument we can find infinitely many squares in the sequence. $\square$

Problem 2. Prove that one can eliminate some terms of an arithmetic sequence of positive integers in such a way that the remaining terms form a geometric sequence.

Solution. Let a be the first term and d the difference of the arithmetic sequence. Observe that one of the terms of the sequence is $a + ad = a(1 + d)$. Another one is $a + (2a + ad)d = a(1 + d)^2$. Hence all the terms in the sequence of the form $a(1 + d)^n$, where $n \geq 1$, form a geometric sequence. $\square$

Problem 3. Does there exist an infinite non-constant arithmetic sequence whose terms are all prime numbers?

Solution. No. Indeed, if such sequence exists, let its common difference be d . Now consider the d consecutive integers

$$(d + 1)! + 2, (d + 1)! + 3, \dots, (d + 1)! + d + 1.$$

Each of them is a composite number, but since the sequence has the common difference d , one out of any d consecutive integers must be a term of the sequence. This is a contradiction. $\square$

Problem 4. If the set of positive integers is partitioned into finitely many arithmetic sequences, prove that at least one of these arithmetic sequences has the property that its first term is divisible by its common difference.

Solution. Let a_i and d_i , where $i = 1, 2, \dots, n$, be the first terms and the common differences of the arithmetic sequences. Now consider the number

$$d = d_1 d_2 \cdot \dots \cdot d_n.$$

This number must belong to one of the arithmetic sequences, thus, for some i and k we have

$$d_1 d_2 \cdot \dots \cdot d_n = a_i + k d_i.$$

It follows that a_i is divisible by d_i . $\square$

Problem 5. If the set of positive integers is partitioned into n arithmetic sequences with common differences $d_1, d_2, \dots, d_n$ prove that

$$\frac{1}{d_1} + \frac{1}{d_2} + \dots + \frac{1}{d_n} = 1.$$

Solution. Let $a_k = a_1 + (k - 1)d_1$ be the sequence with common difference d_1 . Let us count the number of terms of this sequence that are less than or equal to some positive integer N . The inequality $a_k \leq N$ is equivalent to

$$k \leq \frac{N}{d_1} - \frac{a_1}{d_1} + 1.$$

It follows that the number of terms of the sequence with common difference d_1 whose terms belong to the set $\{1, 2, \dots, N\}$ equals

$$\left\lfloor \frac{N}{d_1} - \frac{a_1}{d_1} + 1 \right\rfloor.$$

Similarly, we deduce that the number of terms of the sequence with common difference d_i whose terms belong to the set $\{1, 2, \dots, N\}$ equals

$$\left\lfloor \frac{N}{d_i} - \frac{a_i}{d_i} + 1 \right\rfloor,$$

where a_i is its corresponding first term. Since the sequences partition the set of positive integers, we have

$$\sum_{i=1}^n \left\lfloor \frac{N}{d_i} - \frac{a_i}{d_i} + 1 \right\rfloor = N.$$

Using the inequality $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$, we obtain

$$N \leq \sum_{i=1}^n \left(\frac{N}{d_i} - \frac{a_i}{d_i} + 1 \right) < N + n.$$

Consequently

$$1 \leq \sum_{i=1}^n \frac{1}{d_i} - \frac{1}{N} \sum_{i=1}^n \frac{a_i}{d_i} + \frac{n}{N} < 1 + \frac{n}{N}.$$

Letting $N \rightarrow \infty$ yields the desired result. $\square$

Problem 6 (APMO 2013 Problem 3). For $2k$ real numbers $a_1, a_2, \dots, a_k, b_1, b_2, \dots, b_k$, define the sequence of numbers $\{X_n\}$ by

$$X_n = \sum_{i=1}^k \lfloor a_i n + b_i \rfloor \quad \text{for } n = 1, 2, \dots$$

If the sequence $\{X_n\}$ forms an arithmetic sequence, show that $\sum_{i=1}^k a_i$ must be an integer

Solution. Let $A = \sum_{i=1}^k a_i$ and $B = \sum_{i=1}^k b_i$. Summing the inequalities over i ,

$$a_i n + b_i - 1 < \lfloor a_i n + b_i \rfloor \leq a_i n + b_i,$$

we obtain

$$An + B - k < X_n \leq An + B.$$

Now suppose that $\{X_n\}$ forms an arithmetic sequence with common difference d . Then we have $X_{n+1} = X_1 + nd$ and $A + B - k < X_1 \leq A + B$. Combining the inequalities obtained above, we have

$$An - k \leq An + (A + B - X_1) - k < nd \leq An + (A + B - X_1) < An + k$$

from which we conclude

$$|A - d| < \frac{k}{n}$$

must hold. Since this inequality holds for any positive integer n , we must have $A = d$. Since $\{X_n\}$ is clearly a sequence of integers, d must be an integer also, and therefore A is also an integer. $\square$

Problem 7. Prove that, for every positive integer $n \geq 3$, one can find an arithmetic sequence of n integers such that all of them are nontrivial powers of some integers, but one cannot find an infinite arithmetic sequence with this property.

Solution. We prove the assertion by induction on n . For $n = 3$, we can consider the numbers 1, 25, 49. Suppose that the assertion is true for some $n \geq 3$ and let $a_i = b_i^{k_i}$, for $i = 1, 2, \dots, n$, be the terms of the sequence with common difference d . Let $b = a_n + d$ and let $k = \text{lcm}(k_1, k_2, \dots, k_n)$. Then the $n + 1$ numbers

$$a_1 b^k, a_2 b^k, \dots, a_n b^k, b^{k+1}$$

form an arithmetic sequence and all are powers of some integers. Indeed, for all $i = 1, 2, \dots, n$, there exists d_i such that $k = k_i d_i$ and we obtain

$$a_i b^k = b_i^{k_i} b^k = b_i^{k_i} b^{k_i d_i} = (b_i b^{d_i})^{k_i},$$

as desired.

For the second part, we need the following fact: if $\{a_n\}$ is an arithmetic sequence of positive integers, then

$$\sum_{n=1}^{\infty} \frac{1}{a_n} = +\infty.$$

Now suppose that $\{a_n\}$ is an arithmetic sequence in which all terms are powers of some integers. Let S be the set of all positive integers greater than 1 that are powers of some integers. We will prove that

$$\sum_{a \in S} \frac{1}{a} \leq 1,$$

which clearly contradicts the above-mentioned fact. Indeed, we have

$$\sum_{a \in S} \frac{1}{a} \leq \sum_{n \geq 2} \sum_{k \geq 2} \frac{1}{n^k} = \sum_{n \geq 2} \frac{1}{n^2} \left(1 + \frac{1}{n} + \frac{1}{n^2} + \dots \right) = \sum_{n \geq 2} \frac{1}{n^2} \cdot \frac{1}{1 - \frac{1}{n}} = \sum_{n \geq 2} \frac{1}{n(n-1)} = 1. \quad \square$$

Problem 8. Let $a_1, a_2, a_3, \dots$ be an arithmetic sequence such that a_1^2, a_2^2 and a_3^2 are also terms of the sequence. Prove that the terms of this sequence are all integers.

Solution. Let d be the common difference. If $d = 0$, then either $a_n = 0$ or $a_n = 1$ for all $n = 1, 2, \dots$

Suppose that $d \neq 0$, and consider positive integers m, n, p such that

$$\begin{cases} a_1^2 = a_1 + md, \\ (a_1 + d)^2 = a_1 + nd, \\ (a_1 + 2d)^2 = a_1 + pd. \end{cases}$$

Subtracting the first equation from the other two gives

$$\begin{cases} 2a_1 + d = n - m \\ 4a_1 + 4d = p - m, \end{cases}$$

and solving for a_1 and d , we have $a_1 = \frac{1}{4}(4n - 3m - p)$ and $d = \frac{1}{2}(m - 2n + p)$. So both a_1 and d are rational numbers. Now observe that the equation $a_1^2 = a_1 + md$ can be written as

$$a_1^2 + (2m - 1)a_1 - m(d + 2a_1) = 0,$$

or equivalently,

$$a_1^2 + (2m - 1)a_1 - m(n - m) = 0.$$

This means that a_1 is a root of the monic polynomial $p(x)$ with integer coefficients given by

$$p(x) = x^2 + (2m - 1)x - m(n - m).$$

By the Integer Root Theorem, a_1 is an integer and hence $d = n - m - 2a_1$ is an integer as well. The result follows. $\square$

Problem 9 (China MO 2023 Problem 1). Let $\{a_n\}$ and $\{b_n\}$ be sequences of positive numbers satisfying the following

$$a_{n+1} = a_n - \frac{1}{1 + \sum_{i=1}^n \frac{1}{a_i}}, \quad b_{n+1} = b_n + \frac{1}{1 + \sum_{i=1}^n \frac{1}{b_i}},$$

for $n = 1, 2, \dots$

- (1) If $a_{100}b_{100} = a_{101}b_{101}$, find the value of $a_1 - b_1$.
- (2) If $a_{100} = b_{99}$, determine which one is larger, $a_{100} + b_{100}$ or $a_{101} + b_{101}$?

Solution. (1) Let $a_1 = a$ and $b_1 = b$. From the given conditions, $a, b > 0$ and, for $n \geq 2$, we have

$$\frac{1}{a_n - a_{n+1}} = 1 + \sum_{i=1}^n \frac{1}{a_i} = \frac{1}{a_{n-1} - a_n} + \frac{1}{a_n}.$$

Then we have

$$\frac{a_n}{a_n - a_{n+1}} = \frac{a_n}{a_{n-1} - a_n} + 1 = \frac{a_{n-1}}{a_{n-1} - a_n}$$

from which we have $\frac{a_{n+1}}{a_n} = \frac{a_n}{a_{n-1}}$, and inductively, we obtain

$$\frac{a_{n+1}}{a_n} = \frac{a_n}{a_{n-1}} = \dots = \frac{a_2}{a_1} = \frac{a}{a+1}.$$

In a similar way, we find the ratio $\frac{b_{n+1}}{b_n}$ as follows. From the given conditions, for $n \geq 2$, we have

$$\frac{1}{b_{n+1} - b_n} = 1 + \sum_{i=1}^n \frac{1}{b_i} = \frac{1}{b_n - b_{n-1}} + \frac{1}{b_n}.$$

Then we have

$$\frac{b_n}{b_{n+1} - b_n} = \frac{b_n}{b_n - b_{n-1}} + 1 = \frac{b_{n-1}}{b_n - b_{n-1}} + 2 = \dots = \frac{b_1}{b_2 - b_1} + 2(n-1) = b + (2n-1).$$

Hence $\frac{b_{n+1}}{b_n} = \frac{b+2n}{b+2n-1}$.

Therefore, if $a_{100}b_{100} = a_{101}b_{101}$, then we have

$$\frac{a+1}{a} = \frac{a_{100}}{a_{101}} = \frac{b_{101}}{b_{100}} = \frac{b+200}{b+199},$$

which gives $a = b + 199$, that is, $a_1 - b_1 = a - b = 199$.

(2) Note that $\{a_n\}$ is a strictly decreasing sequence, while $\{b_n\}$ is a strictly increasing sequence, and since $a_{100} = b_{99}$, we must have

$$a_1 > a_2 > \dots > a_{99} > a_{100} = b_{99} > b_{98} > \dots > b_1 > 0.$$

Hence

$$a_{99} = a_{100} + \frac{1}{1 + \sum_{i=1}^{99} \frac{1}{a_i}} > b_{99} + \frac{1}{1 + \sum_{i=1}^{99} \frac{1}{b_i}} = b_{100}.$$

Consequently

$$a_{100} - a_{101} = \frac{1}{1 + \sum_{i=1}^{98} \frac{1}{a_i} + \frac{1}{a_{99}} + \frac{1}{a_{100}}} > \frac{1}{1 + \sum_{i=1}^{98} \frac{1}{b_i} + \frac{1}{b_{100}} + \frac{1}{b_{99}}} = b_{101} - b_{100}.$$

Therefore $a_{100} + b_{100} > a_{101} + b_{101}$. □

Problem 10 (IMO 2009 Problem 3). Suppose that $s_1, s_2, s_3, \dots$ is a strictly increasing sequence of positive integers such that the subsequences

$$s_{s_1}, s_{s_2}, s_{s_3}, \dots \quad \text{and} \quad s_{s_1+1}, s_{s_2+1}, s_{s_3+1}, \dots$$

are both arithmetic sequences. Prove that $s_1, s_2, s_3, \dots$ is itself an arithmetic sequence.

Solution. From the given condition, we know that $s_{s_1}, s_{s_2}, s_{s_3}, \dots$ and $s_{s_1+1}, s_{s_2+1}, s_{s_3+1}, \dots$ are both strictly increasing sequences of positive integers. Let

$$s_{s_k} = a + (k-1)d_1 \quad \text{and} \quad s_{s_k+1} = b + (k-1)d_2,$$

where $k = 1, 2, \dots$, and a, d_1, b, d_2 are positive integers.

Since $s_k < s_k + 1 \leq s_{k+1}$ and by the monotonicity of $\{s_n\}$, we have

$$s_{s_k} < s_{s_k+1} \leq s_{s_{k+1}}$$

for $k = 1, 2, \dots$. Hence

$$a + (k-1)d_1 < b + (k-1)d_2 \leq a + kd_1,$$

or equivalently,

$$a - b < (k-1)(d_2 - d_1) \leq a - b + d_1.$$

Since this inequality holds for any positive integer k , we have $d_2 - d_1 = 0$. Now, let $d = d_1 = d_2$ and $b - a = c$. Note that c is a positive integer.

If $d = 1$, then by the monotonicity of $\{s_n\}$, we have $s_{s_{k+1}} = s_{s_k} + 1 \leq s_{s_{k+1}}$, and so $s_{k+1} \leq s_k + 1$. Since $s_k < s_{k+1}$, we have $s_{k+1} = s_k + 1$, which means that $\{s_n\}$ is an arithmetic sequence.

If $d > 1$, we are going to show that, for any positive integer k , we have $s_{k+1} - s_k = c$.

CASE 1: Suppose that there exists a positive integer k such that $s_{k+1} - s_k < c$. Since $s_{k+1} - s_k$ takes only positive integer values, it must have a minimum value, say $s_{i+1} - s_i = c_0$. Then

$$\begin{aligned} s_{a+id} - s_{a+(i-1)d+1} &= s_{s_{s_{i+1}}} - s_{s_{s_i}+1} \\ &= (a + (s_{i+1} - 1)d - (b + (s_i - 1)d)) \\ &= c_0d - c. \end{aligned}$$

On the other hand, since

$$(a + id) - (a + (i-1)d + 1) = d - 1,$$

we have

$$s_{a+id} - s_{a+(i-1)d+1} \geq c_0(d-1)$$

which implies that $c_0 \geq c$, a contradiction.

CASE 2: Suppose that there exists a positive integer k such that $s_{k+1} - s_k > c$. Since $s_{k+1} - s_k$ takes only integer values, and for each given k ,

$$s_{k+1} - s_k \leq s_{s_{k+1}} - s_{s_k} = d,$$

it must have a maximum value, say $s_{j+1} - s_j = c_1$. Then

$$\begin{aligned} s_{a+jd} - s_{a+(j-1)d+1} &= s_{s_{s_{j+1}}} - s_{s_{s_j}+1} \\ &= (a + (s_{j+1} - 1)d - (b + (s_j - 1)d)) \\ &= c_1d - c. \end{aligned}$$

On the other hand, since

$$(a + jd) - (a + (j - 1)d + 1) = d - 1,$$

we have

$$s_{a+jd} - s_{a+(j-1)d+1} \leq c_1(d - 1)$$

which implies that $c_1 \leq c$, a contradiction.

Therefore, for any positive integer k , we have $s_{k+1} - s_k = c$ and the result follows. $\square$

Problem 11 (IMO 2003 Problem 5). Let $x_1, x_2, \dots, x_n$ be real numbers such that $x_1 \leq x_2 \leq \dots \leq x_n$. Prove that

$$\left(\sum_{i=1}^n \sum_{j=1}^n |x_i - x_j| \right)^2 \leq \frac{2(n^2 - 1)}{3} \sum_{i=1}^n \sum_{j=1}^n (x_i - x_j)^2,$$

where equality holds if and only if $x_1, x_2, \dots, x_n$ form an arithmetic sequence.

Solution. Since both sides of the inequality are invariant under any translation of all x_i 's, we may assume, without loss of generality that $\sum_{i=1}^n x_i = 0$.

Observe that

$$\sum_{i=1}^n \sum_{j=1}^n |x_i - x_j| = 2 \sum_{1 \leq i < j \leq n} (x_j - x_i) = 2 \sum_{i=1}^n (i - 1 - (n - i))x_i = 2 \sum_{i=1}^n (2i - n - 1)x_i.$$

By the Cauchy-Schwarz Inequality, we have

$$\left(\sum_{i=1}^n \sum_{j=1}^n |x_i - x_j| \right)^2 \leq 4 \sum_{i=1}^n (2i - n - 1)^2 \sum_{i=1}^n x_i^2 = 4 \cdot \frac{n(n^2 - 1)}{3} \sum_{i=1}^n x_i^2.$$

On the other hand, we have

$$\sum_{i=1}^n \sum_{j=1}^n (x_i - x_j)^2 = n \sum_{i=1}^n x_i^2 - \sum_{i=1}^n x_i \sum_{j=1}^n x_j + n \sum_{j=1}^n x_j^2 = 2n \sum_{i=1}^n x_i^2.$$

Therefore

$$\left(\sum_{i=1}^n \sum_{j=1}^n |x_i - x_j| \right)^2 \leq \frac{4n(n^2 - 1)}{3} \sum_{i=1}^n x_i^2 = \frac{4n(n^2 - 1)}{3} \cdot \frac{1}{2n} \sum_{i=1}^n \sum_{j=1}^n (x_i - x_j)^2 = \frac{2(n^2 - 1)}{3} \sum_{i=1}^n \sum_{j=1}^n (x_i - x_j)^2.$$

If the equality holds, then $x_i = k(2i - n - 1)$ for some real number k , which means that $x_1, x_2, \dots, x_n$ form an arithmetic sequence.

On the other hand, suppose that $x_1, x_2, \dots, x_n$ form an arithmetic sequence with common difference d . Then we have

$$x_i = \frac{d}{2}(2i - n - 1) + \frac{1}{2}(x_1 + x_n).$$

Translate the x_i 's by $-\frac{1}{2}(x_1 + x_n)$ to obtain $x_i = \frac{d}{2}(2i - n - 1)$ with $\sum_{i=1}^n x_i = 0$ from which the inequality follows. $\square$

Problem 12 (China TST 2019 Test 1 Problem 2). Fix a positive integer $n \geq 3$. Does there exist infinitely many sets S of $2n$ positive integers $\{a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n\}$, such that $\gcd(a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n) = 1$, and $\{a_1, a_2, \dots, a_n\}, \{b_1, b_2, \dots, b_n\}$ are arithmetic sequences with $\prod_{i=1}^n a_i = \prod_{i=1}^n b_i$?

Solution. Yes, there exist infinitely many such sets.

Let $a_k = r(a + (k-1)d)$, $b_k = s(a + kd)$, for $k = 1, 2, \dots, n$, where a, d, r, s are positive integers to be determined below. Note that $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ are strictly increasing arithmetic sequences.

Now observe that the condition $\prod_{i=1}^n a_i = \prod_{i=1}^n b_i$ is equivalent to $r^n a = s^n(a + nd)$, that is,

$$(r^n - s^n)a = s^n dn. \quad (*)$$

Below we choose a, d, r, s satisfying $(*)$ such that $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$ are pairwise distinct and with greatest common divisor equal to 1.

For any positive integer t , let $a = s = 1$, $r = tn + 1$, and $d = \frac{r^n - 1}{n} = \frac{(tn + 1)^n - 1}{n}$. Then, clearly, $r > n$, d is a positive integer, and $(*)$ is satisfied.

Since $\gcd(b_1, b_2) = \gcd(1+d, 1+2d) = \gcd(1+d, d) = 1$, it follows that $\gcd(a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n) = 1$. Now if $a_i = b_j$ for some $i, j \in \{1, 2, \dots, n\}$, then

$$r(n + (i-1)(r^n - 1)) = na_i = nb_j = n + j(r^n - 1) \quad (**)$$

Taking mod r , we have $0 \equiv n - j \pmod{r}$. Since $r = tn + 1 > n$, we must have $j = n$ and so $(**)$ becomes

$$n + (i-1)(r^n - 1) = nr^{n-1}.$$

Taking mod r again, we have $n - i + 1 \equiv 0 \pmod{r}$. Since $0 < n - i + 1 \leq n < r$, it follows that r cannot divide $n - i + 1$. So $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$ are pairwise distinct and satisfy all the problem conditions. $\square$

Problem 13 (USAMO 2023 Problem 5). Let $n \geq 3$ be an integer. We say that an arrangement of the numbers $1, 2, \dots, n^2$ in a $n \times n$ table is *row-valid* if the numbers in each row can be permuted to form an arithmetic sequence, and *column-valid* if the numbers in each column can be permuted to form an arithmetic sequence.

For what values of n is it possible to transform any row-valid arrangement into a column-valid arrangement by permuting the numbers in each row?

Solution. The answer is that the transformation is always possible when n is prime, but not always when n is composite.

Suppose first that n is prime, and let C be any $n \times n$ row-valid table. Observe that, for each remainder mod n , there are exactly n positive integers up to n^2 that leave that remainder mod n . Let us call a row of C *uniform* if its n elements all leave the same remainder mod n , and *diverse* if the n elements all leave a different remainder mod n .

It is easy to see that each row of C is either uniform or diverse. Indeed, if we had two integers in a row that left the same remainder mod n , then their difference, which is a multiple of the common difference of the arithmetic sequence corresponding to that row, would be a multiple of n , and thus all elements in that row would leave the same remainder mod n . Observe also that if C contains a uniform row, then that row includes all numbers with that remainder, so C cannot also contain a diverse row.

Therefore, we have two cases for our row-valid table: either every row is uniform or every row is diverse. In the first case, we can permute the elements in each row so that the j th column contains the elements of the arithmetic sequence

$$1 + (j-1)n, 2 + (j-1)n, \dots, n + (j-1)n$$

in some order, while in the second case, we can arrange that the j th column contains the arithmetic sequence

$$j, j + n, j + 2n, \dots, j + (n - 1)n$$

in some order. This proves our claim when n is prime.

Suppose now that n is composite with a proper divisor $c > 1$. To construct an example for a row-valid table A that cannot be transformed into a column-valid one, we arrange the first n^2 positive integers by setting the j th entry in row i to be

$$a_{i,j} = \begin{cases} i + (j - 1)c & \text{if } i \leq c, \\ j + (i - 1)n & \text{if } i > c, \end{cases}$$

(Thus the first cn positive integers occupy the first c rows of A , arranged in order by columns, and the rest of the integers are placed in the last $n - c$ rows, arranged in order by rows. An illustration for $n = 9$ and $c = 3$ is provided below.)

1	4	7	10	13	16	19	22	25
2	5	8	11	14	17	20	23	26
3	6	9	12	15	18	21	24	27
28	29	30	31	32	33	34	35	36
37	38	39	40	41	42	43	44	45
46	47	48	49	50	51	52	53	54
55	56	57	58	59	60	61	62	63
64	65	66	67	68	69	70	71	72
73	74	75	76	77	78	79	80	81

Clearly, each row in A is an arithmetic sequence. Suppose that it is possible to permute the elements in the rows so that the table becomes column-valid; let table B be the result of these permutations. Consider the column of B that contains 2. The elements in that column cannot contain 1, since the arithmetic sequence then would be $(1, 2, \dots, n)$, which includes $2 + c$ (since $2 + c \leq n$ for $n \geq 4$), but 2 and $2 + c$ are both in the second row (since $2 + c = a_{2,2}$).

Therefore, the elements in B in the column containing 2 are $\{2 + kd : k = 0, 1, \dots, n - 1\}$ for some positive integer d . The largest element in this set is $2 + (n - 1)d$, for which we must have

$$n^2 - n + 1 \leq 2 + (n - 1)d \leq n^2$$

as this element is in the last row of A ; and these inequalities yield $d = n$. We have arrived at a contradiction, since the integer $2 + n$ is then in the column containing 2 (with $k = 1$), but $2 + n$ is in the same row as 2 (since $a_{2,j} = 2 + n$ for $j = \frac{n}{c} + 1$). This is impossible. $\square$